

Poster: GEM-PLI Side Channel in GPON

Inferring Encrypted Streaming Traffic

Hyunjun An
Heungdeok High School
hjahn@anhyunjun.com

Abstract—Gigabit Passive Optical Networks (GPON), an ITU-T standardized technology widely adopted in fiber-to-the-home (FTTH) networks, supports downstream payload encryption to protect subscribers' traffic. Unfortunately, encryption does not cover all access-layer metadata. This paper presents a preliminary analysis showing that the Payload Length Indicator (PLI) which remains in plaintext in GEM headers forms a side channel that enables inference of streaming activity. A passive attacker on the same optical splitter as victim can observe downstream PLI and infer victim's traffic streaming activity (e.g., Inferring Youtube Video traffic identification).

I. INTRODUCTION AND MOTIVATION

As fiber internet access has become dominant in Network Infrastructures, GPON—one of the passive optical network (PON) technologies that is widely adopted by many ISPs to provide Fiber Internet for home (FTTH) and Enterprise—accounted for 54.2% of FTTH market in 2024 [5]. Since GPON is popular PON technology, it is important to ensure traffic confidentiality. GPON supports AES-based downstream traffic encryption on GEM Layer. However, some fields of GEM Header remain plaintext such as the Payload Length Indicator (PLI) field used for multiplexing and the Port-ID field used for identifying subscribers' Optical Network Terminals/Units (ONTs/ONUs). Such plaintext fields may appear innocuous. However, recent studies have demonstrated that such metadata can be used to infer victim's activity (e.g., identifying a victim's web-based video streaming watching activity) [2] [6] [7]. Prior studies primarily focused on upper-layer or cellular traffic analysis [2] [6]; in contrast optical networks, particularly PON networks which is primarily used for Fiber Internet for home have received no attention. Inspired by prior studies, we researched whether the plaintext PLI field in GPON GEM header can be used to leak victim's privacy on the same optical splitter. We conduct a preliminary experimental analysis using a software-based GPON downstream simulator to evaluate whether the plaintext PLI field can be exploited to infer victim activity. Using these PLI traces, we present preliminary first evidence that Youtube streaming video IDs can be identified by attacker in GPON networks.

II. BACKGROUND

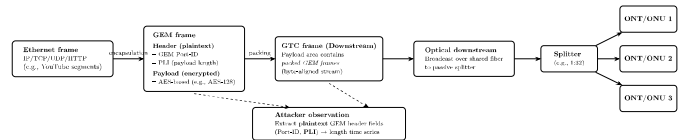


Fig. 1. Encapsulation pipeline in GPON downstream: upper-layer packets (e.g., Ethernet/IP) are encapsulated into GEM frames, packed into GTC frames, and broadcast over the shared optical line. While GEM payloads are AES encrypted (e.g., AES-128), GEM header fields such as Port-ID and PLI remain plaintext.

GPON Topology and GTC Framing GPON Networks consist of an Optical Line Terminal (OLT) at the ISP and multiple Optical Network Terminals/Units (ONTs/ONUs) (typically called "modem") deployed at subscribers. A single optical tree can serve up to 32 subscribers (e.g., 1:32 split). In the downstream, the OLT transmits downstream traffic through the splitter. Since the splitter passively distributes the optical signal from the OLT to all ONUs/ONTs on the same splitter, each ONU/ONT can physically receive the downstream transmissions destined for other ONUs/ONTs. At the transmission layer, downstream bytes are carried in GPON Transmission Convergence (GTC) frames sent periodically (per-125 μ s), whose payload area carries multiple GEM frames (described next).

GEM Framing In GPON Network, upper layer traffic (e.g., Ethernet frames carrying TCP/IP packets) in the GEM Payload Field transmitted via GPON Encapsulation Method (GEM) Frames. Downstream GEM payloads are often encrypted for confidentiality (depending on ISP configuration [1]). However, the GEM Port-ID (used to demultiplex services/flows) and the Payload Length Indicator (PLI), which encodes the payload size, are transmitted in plaintext. Thus, an attacker can extract a victim's encrypted payload lengths from consecutive PLI values (optionally grouped by Port-ID), even without decrypting the payload contents.

Video Identification HTTP Adaptive Streaming (HAS) is the popular video streaming model – which is generally used by video streaming platforms (e.g., Youtube, many OTT Services) – delivers video by letting the client periodically request short media segments at a bitrate chosen by an adaptive bitrate (ABR) algorithm. This request–response process forms characteristic bursty download patterns and bitrate-dependent segment sizes, which can be exploited for traffic inference. [2]

III. THREAT MODEL

We consider an attacker on the same optical splitter as a victim.

- **1. Premise:** Attacker should have optical capture device (e.g., FPGA, modified ONU that can receive raw GTC frames) directly connecting to splitter and passively records downstream from OLT. *attacker does not have to get their ONU/ONT linked with OLT so if attacker remove Tx function of optic module, they can entirely record downstream records without being exposed.
- **2. Collect GEMs** Then, attacker extract GTC frames from downstream traffic and they get many GEM frames.
- **3. Convert GEM PLI to dataset** Then, attacker filters GEM frames corresponding to a victim's Port-ID and made a dataset containing PLI values and timestamps.
- **4. Dataset format** The dataset must include start & end time and duration and PLI and sequence information.

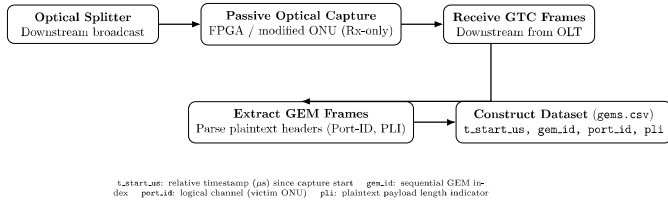


Fig. 2. Attacker-side observation pipeline: passive optical capture under the same splitter enables reception of downstream GTC frames, extraction of GEM frames, and construction of a PLI-based dataset (`gems.csv`).

IV. EXPERIMENT AND PRELIMINARY RESULTS

To demonstrate threat model in our experimental environment, we developed a software GPON downstream simulator that reproduces the process of receiving GTC frames, extracting GEM frames, and exporting plaintext GEM header fields into datasets. At first we played each of 10 videos three times in our linux based system to produce packet capture datas (PCAPs) where other streaming traffic(e.g., VNC) is restricted to minimize interference. then we simulated that PCAPs packet capture file to records the resulting PLI values and timestamps into CSV traces. These traces approximate the payload length observations available to a real same-splitter adversary and enable systematic evaluation of PLI-based inference. After making dataset, we visualized cumulative PLI - Time Graph based on our dataset(Publicly Available on our github repository) [4]. In conclusion, We could find a clear correlation between video and cumulative PLI over time graph; Traces generated from the same video show greatly consistent slopes, which are significantly distinctive to distinguish individual videos. we assume that when adopted CNN, it is possible to distinguish video Ids using captured PLI values.

V. CONCLUSION AND DISCUSSION

This poster show that payload encryption in GEM is insufficient to ensure traffic confidentiality in GPON networks. Despite Payload Encryption, GPON Network is still vulnerable to

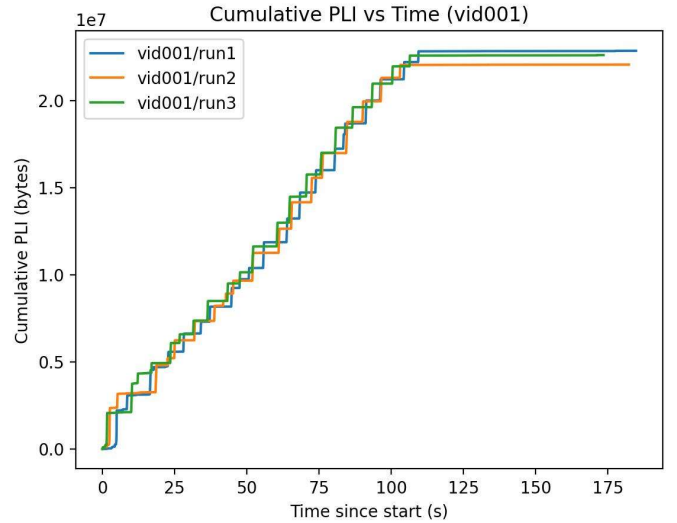


Fig. 3. PLI - Time Graph of one of 10 videos showing that greatly identical slopes between three runs, PLI - Time Graph of others is publicly available on our github repository [4].

inferring streaming activity because PLI in GEM headers can be exploited to infer a victim's video activity(e.g., Youtube). Future work includes validating our observations on a real GPON testbed. All code and experimental dataset is publicly available on the our github repository. [4]

Future Work In Real World. Future Work will be conducted in experimental testbed with Real OLT, ONU and Optical Splitter. We will use FPGA to collect GEM frames.

Extent to XGS-PON Networks. Although we entirely focus on GPON in this poster, we believe that XGS-PON Networks are subject to analogous privacy risks. future work includes XGS-PON.

Open Discussion We welcome open discussion on this poster. if you would like to contact us, please email us.

VI. ACKNOWLEDGMENT

Note: High school student authors—please excuse any imperfect wording.

REFERENCES

- [1] ITU-T Recommendation G.984.3, *Gigabit-capable passive optical networks (GPON): Transmission convergence layer specification*.
- [2] S. Bae KAIST, "Watching the Watchers: Practical Video Identification Attacks in LTE Networks," in *USENIX Security*, 2022.
- [3] Cisco, "Understand GPON Technology," Troubleshooting TechNotes. (Document ID: 216230) -
- [4] github repository - <https://github.com/hjaka08/GPON>
- [5] Fiber-to-the-Home (FTTH) Market Size & Share Analysis - Growth Trends and Forecast (2025 - 2030) - <https://www.mordorintelligence.com/industry-reports/fiber-to-the-home-ftth-market>
- [6] Traffic spills the beans- A robust video identification attack against YouTube - Elsevier Computers & Security (2024)
- [7] Breaking Through the Diversity: Encrypted Video Identification Attack Based on QUIC Features - ESORICS 2024

POSTER: GEM-PLI SIDE CHANNEL IN GPON INFERRING ENCRYPTED STREAMING TRAFFIC

HyunJun An

Heungdeok High School
hjahn@anhyunjun.com

INTRODUCTION & MOTIVATION

GPON – one of the passive optical network (PON) technologies-accounted for 54.2% of Fiber To The Home (FTTH) market in 2024

it is important to ensure traffic confidentiality.

2. GPON supports AES-based downstream traffic encryption on GEM Layer.

3. However, some fields of GEM Header remain plaintext

4. recent studies have demonstrated that such metadata like PLI can be used to infer victim's activity.

5. we researched whether the plaintext PLI field in GPON GEM header can be used to leak victim's privacy on the same optical splitter

BACKGROUND

GPON Topology and GTC Framing GPON Networks consist of an Optical Line Terminal (OLT) at the ISP and multiple Optical Network Terminals/Units(ONTs/ONUs) (typically called "modem") deployed at subscribers. A single optical tree can serve up to 32 subscribers (e.g., 1:32 split). In the downstream, the OLT transmits downstream traffic through the splitter. Since the splitter passively distributes the optical signal from the OLT to all ONUs/ONTs on the same splitter, each ONU/ONT can physically receive the downstream transmissions destined for other ONUs/ONTs. At the transmission layer, downstream bytes are carried in GPON Transmission Convergence (GTC) frames sent periodically(per-125 μ s), whose payload area carries multiple GEM frames(described next).

METHODOLOGY

LWe consider a attacker on the same optical splitter as a victim.

- 1. Premise: Attacker should have optical capture device (e.g., FPGA, modified ONU that can receive raw GTC frames) directly connecting to splitter and passively records downstream from OLT. *attacker does not have to get their ONU/ONT linked with OLT so if attacker remove Tx function of optic module, they can entirely record downstream records without being exposed.
- 2. Collect GEMs Then, attacker extract GTC frames from downstream traffic and they get many GEM frames.
- 3. Convert GEM PLI to dataset Then, attacker filters GEM frames corresponding to a victim's Port-ID and made a dataset containing PLI values and timestamps. *
- 4. Dataset format The dataset must include start & end time and duration and PLI and sequence information.

EXPERIMENT

we developed a software GPON downstream simulator that reproduces the process of receiving GTC frames, extracting GEM frames, and exporting plaintext GEM header fields into datasets. At first we played each of 10 videos three times in our linux based system to produce packet capture datas (PCAPs) where other streaming traffic(e.g., VNC) is restricted to minimize interference. then we simulated that PCAPs packet capture file to records the resulting PLI values and timestamps into CSV traces

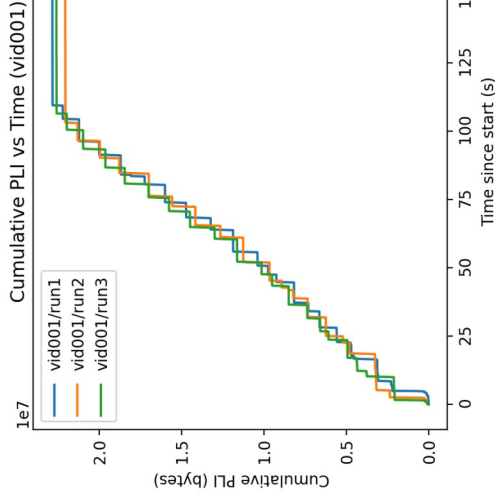


fig. PLI - Time Graph for one of 10 videos showing that greatly identical slopes between three runs, PLI - Time Graph of others is publicly available on our github repository

CONCLUSION

GPON Network is vulnerable to streaming inferring attack.
Payload encryption alone does not ensure confidentiality of traffic.

FUTURE WORK

Future Work will be conducted in experimental testbed with Real OLT and Optical Splitter. We will use FPGA to collect GEM frames. when adopted CNN, it is possible to distinguish video lds using captured PLI values.

DATASET & CODES

Our datasets and codes are publicly available on our github
<https://github.com/hjaka08/GPON>

ACKNOWLEDGEMENTS

I'm a Korean High School Student, please excuse imperfect wording.
Thank you!

REFERENCES

- [1] ITU-T Recommendation G.984.3, Gigabit-capable passive optical networks (GPON): Transmission convergence layer specification.
- [2] S. Bae KAIST, "Watching the Watchers: Practical Video Identification Attacks in LTE Networks," in USENIX Security, 2022.
- [3] Cisco, "Understand GPON Technology," Troubleshooting TechNotes.
- [4] github repository - <https://github.com/hjaka08/GPON>
- [5] Fiber-to-the-Home (FTTH) Market Size & Share Analysis - Growth Trends and Forecast (2025 - 2030)
- [6] Traffic spills the beans: A robust video identification attack against YouTube - Elsevier Computers & Security (2024)
- [7] Breaking Through the Diversity: Encrypted Video Identification Attack Based on QUIC Features - ESORICS 2024